Predicting Indoor Location from Wi-Fi Signal Strength: A Phase 1 Baseline on UJIIndoorLoc

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Project 9 — Applied Machine Learning

Abstract - We estimate where a phone is - which building, which floor, and its coordinates - from Wi-Fi signal strengths. On UJIIndoorLoc, we build separate tree models for each target using the supplied train/validation split. Random Forest reads the building almost perfectly (99.9%) and gives the tightest coordinates (6.86 m median); XGBoost is the better floor classifier (92.0%). Routing each task to whichever model wins it yields a post-hoc combined baseline at 11.58 m mean / 6.86 m median Euclidean positioning error (11.95 m mean / 7.00 m median under our penalized combined metric). Wrong-floor cases have larger coordinate errors, and errors vary more than three-fold across handsets. These results set the bar for a proposed shared multitask network.

Index Terms— indoor localization, Wi-Fi fingerprinting, UJIIndoorLoc, Random Forest, XGBoost, multitask learning.

I. INTRODUCTION

Satellite positioning stops working once you walk indoors, so phones fall back on whatever signals are around — and Wi-Fi is everywhere. The idea behind fingerprinting is simple: record, at many known spots, how loud each nearby access point sounds, then match a new reading against that map. UJIIndoorLoc is the standard benchmark for this. It covers three university buildings with several floors each, and every scan lists the received signal strength (RSSI) of up to 520 access points.

What makes the problem interesting is that it is really three problems stacked together. You want the building (a 3-way choice), the floor (5 possibilities), and a pair of coordinates in metres. These are not independent — knowing the building narrows the floors, knowing the floor constrains the coordinates. This report is Phase 1: a classical machine-learning baseline that treats each of the three as its own model and measures how far that gets us. It is meant to be honest and reproducible rather than clever. Phase 2 will replace it with a network that learns the three targets jointly; the point of a baseline is to know what "jointly" has to improve on.

II. DATA AND EXPLORATION

The training file holds 19,937 scans and the separate validation file holds 1,111. We use that split as given. Because the validation scans were gathered later and on different phones, it is a genuine test of generalisation rather than a reshuffle of the same recordings, and we never fit any transform on it.

A first cleaning pass found 637 training scans that were exact copies of another row — same signals, same labels — so we dropped them and kept 19,300 unique scans. The first thing exploration makes obvious is how little each scan actually sees: the median fingerprint picks up only 17 of the 520 access points (Fig. 1), and 96.5% of the whole signal matrix is empty. Fifty-five access points are never heard anywhere in training, so they carry no information and we drop their columns.

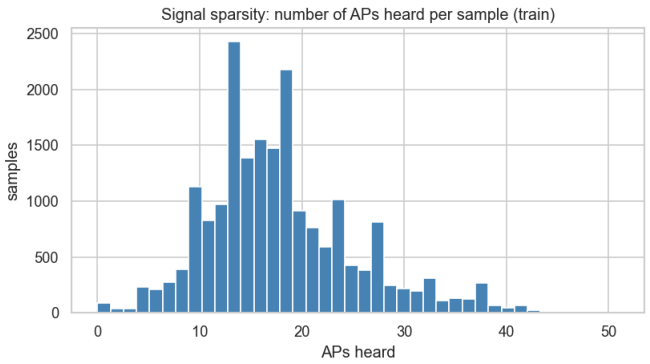


Fig. 1. How many access points a single scan detects. Most see fewer than twenty of the 520 available — the data is sparse by nature.

The three buildings sit in clearly separate places on the map (Fig. 2), which is a useful hint early on: telling the buildings apart should be easy, and the real difficulty will be floors and fine positioning. The floors are not evenly represented. Floor 4 exists only in one building and shows up in roughly a thousand scans, far fewer than the others, so we watch it carefully during evaluation.

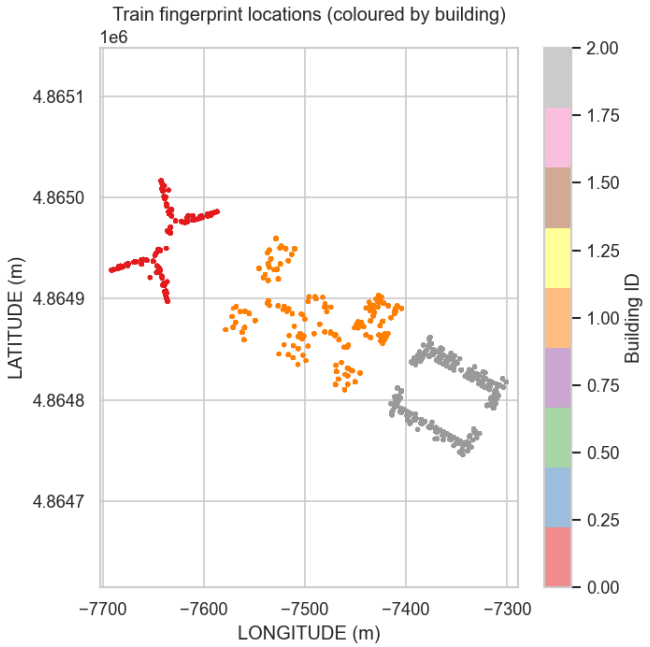


Fig. 2. Training fingerprints at their true coordinates, coloured by building. The three buildings occupy distinct regions.

III. METHOD

Preparing the signals. UJIIndoorLoc marks an access point that was not heard with the value +100, which is not a real signal level and would wreck any distance or scaling if left alone. We remap it to −105 dBm, just below the weakest genuine reading (−104), so "not

heard” becomes a consistent floor value. The observed RSSI is then standardised using statistics from the training data only, and the same transform is applied unchanged to the validation set — so the validation data is free to drift rather than being forced to look normalised.

Features. On top of the 465 surviving RSSI columns we add 465 binary absence flags, one per access point. Standard scaling is invertible: weak-but-heard and absent RSSI remain distinct. Trees use axis-aligned splits; binary flags explicitly expose AP visibility through a boolean threshold. They are a representation choice, not proof of saved tree depth or improved accuracy. We also compute per-scan summaries: how many access points were heard, the strongest reading and its AP, and the mean, spread and range of heard signals. The design matrix has 936 columns. Everything here is cheap to compute and easy to explain.

Models. For each of the three targets we train both a Random Forest and an XGBoost model: classifiers for building and floor, regressors for the two coordinates (predicted together). That is six estimators in total, each ignorant of the others - which is exactly the gap Phase 2 is meant to close. Hyperparameters are left at sensible fixed values; we are not tuning a competition entry, we are drawing a baseline.

IV. RESULTS

Classification and position. Table I gives the full picture. Random Forest misses 1 building label in 1,111 scans. XGBoost reaches 91.99% floor accuracy against RF's 90.73%, while RF gives tighter coordinates (6.86 m median versus 10.06 m). Routing building and coordinates from RF and floor from XGBoost gives 11.58 m mean / 6.86 m median Euclidean error, with P90 24.99 m. The penalized score is 11.95 m mean / 7.00 m median. It adds 4 m for a wrong floor and 50 m for a wrong building, once each. This combination is selected post-hoc on the same validation set, not a fresh test.

TABLE I. VALIDATION RESULTS. COMBINED ROUTES TASKS POST-HOC.

Model	Build. acc.	Floor acc.	Coord. mean (m)	Coord. med. (m)
Random Forest	0.9991	0.9073	11.58	6.86
XGBoost	0.9973	0.9199	14.52	10.06
Combined	**0.9991**	**0.9199**	**11.58**	**6.86**

In addition to 2D Euclidean distance, we explicitly compute coordinate MAE. RF achieves 8.19 m on Longitude and 6.80 m on Latitude (7.49 m average MAE), substantially outperforming XGBoost's 9.82 m and 8.54 m (9.18 m average MAE).

The floor confusion matrix (Fig. 3) shows that most errors are off-by-one, on the floor directly above or below the true floor. This is consistent with signal propagation between adjacent floors, but does not establish its cause. The rare Floor 4 is recalled 79.5% of the time by XGBoost and 71.8% by RF. No verified class-reweighting result is available for this baseline.

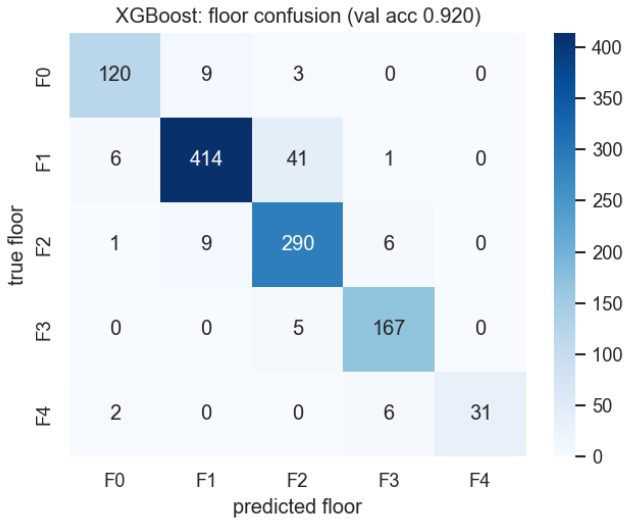


Fig. 3. Floor confusion for XGBoost. Mistakes cluster on the diagonal neighbours — almost every error is a single floor up or down.

Where the error lives. Plotting the coordinate error at each true location (Fig. 4) shows it is not spread evenly. Corridors with dense fingerprint coverage are accurate; the large errors sit at building edges and corners where fewer access points reach. The error CDF (Fig. 5) makes the model comparison clean — Random Forest’s curve sits to the left of XGBoost’s at every percentile, so it is not just better on average but better across the whole distribution of coordinate errors.

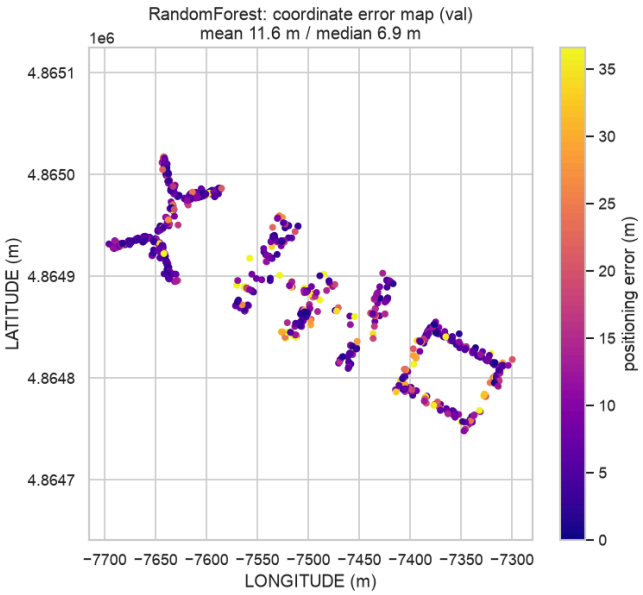


Fig. 4. Coordinate error at each validation point. Error concentrates at building perimeters where coverage thins out.

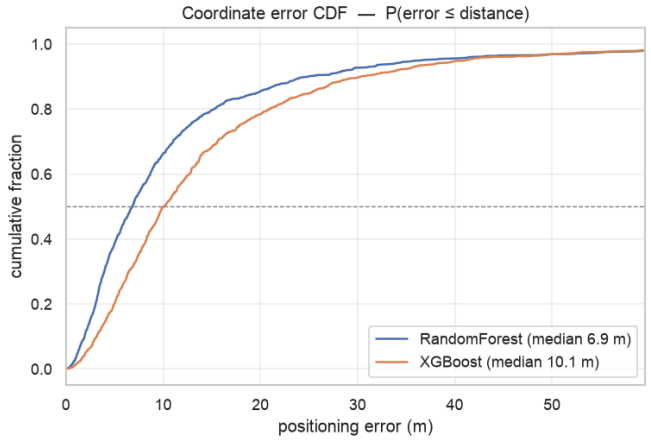


Fig. 5. Cumulative distribution of coordinate error. Random Forest dominates at every distance.

One number ties the tasks together. With RF floor predictions, mean RF coordinate error is 10.19 m when the floor is correct and 25.21 m when it is wrong. This is an association, not evidence that a classification error causes regression error: the separate estimators do not pass predictions to each other. Shared learning is a Phase 2 hypothesis to test.

Robustness across devices. Averages hide a lot. The validation set uses eleven phone models, of which two (Phones 13 and 14, comprising 397 scans) were seen in training, while nine (714 scans) are entirely unseen. Fig. 6 shows RF mean errors from under 8 m to about 31 m, roughly four to one. One phone is poor for both models. XGBoost, while less accurate on average, is a little more even across devices. Device sensitivity matters for deployment and is not revealed by one mean error.

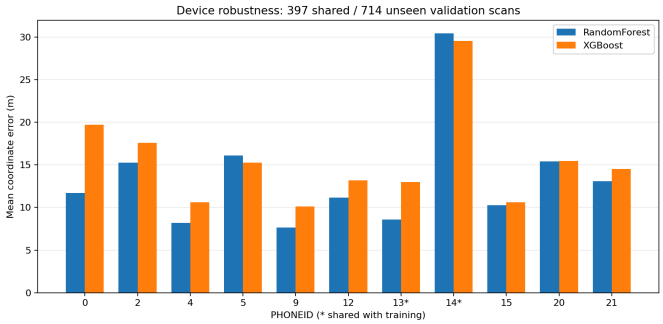


Fig. 6. Mean coordinate error per phone model across validation (397 samples from shared handsets 13 & 14, 714 from nine unseen handsets). Accuracy varies more than three-fold across handsets.

V. DISCUSSION AND NEXT STEPS

The baseline does its job: building is essentially solved, floors land in the low nineties, and median positioning sits around 7 m, from models that are easy to reason about and quick to retrain. We report on one supplied split, not repeated folds, so the figures are point estimates, not confidence intervals. We did no hyperparameter search. Saved USERID-grouped folds were created but are not consumed by the current training script. The three target models never talk to each other.

That last point is the hand-off to Phase 2. The association between floor and coordinate errors, and uneven behaviour across devices, motivate testing a shared representation: one trunk feeding building, floor and coordinate heads. Whether it helps remains to be measured. We will keep the same preprocessing, features, split and metrics for a direct comparison.

REFERENCES

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TEAM CONTRIBUTIONS

Aviral Mishra (M1) led data ingestion, quality audits, deduplication, and USERID-grouped fold creation. Ved Pawar (M2) designed missing-signal remapping, feature representations, and kNN ablations. Khuswant Rajpurohit (M3) trained building/floor classifiers. Prataya Silla (M4) implemented coordinate regressors and spatial diagnostics. Vansh Jain (M5) integrated unified metrics and device robustness evaluations.